

UNSTRING

UNSTRING Syntax.

UNSTRING SourceStr\$i

[DELIMITED BY [ALL] DelimSil [OR [ALL] DelimSil] ...]

INTO {DestStr\$i [DELIMITER IN HoldDelimSi]

[COUNT IN CharCounter#i]} ...

[WITH POINTER Pointer#i]

[TALLYING IN DestCounter#i]

[ON OVERFLOW StatementBlock]

[NOT ON OVERFLOW StatementBlock]

[END-UNSTRING]

```
UNSTRING FullName DELIMITED BY ALL SPACES
      INTO FirstName, SecondName, Surname
END-UNSTRING
```

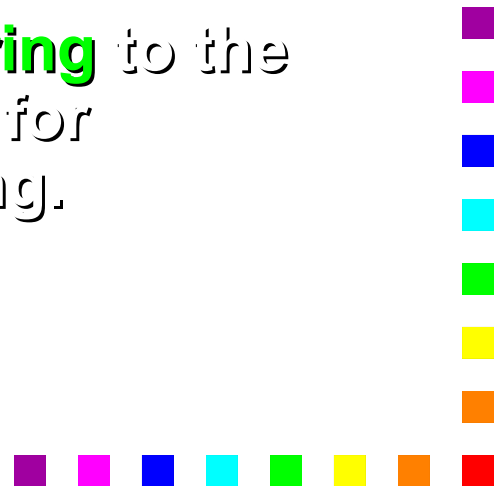
```
UNSTRING CustAddress DELIMITED BY ", "
      INTO Adrline(1), Adrline(2), Adrline(3),
           Adrline(4), Adrline(5), Adrline(6)
      TALLYING IN AdrlinesUsed
END-UNSTRING.
```

How the UNSTRING works

The UNSTRING copies characters from the **Source String** to the **Destination String** until a **Delimiter** is encountered in the **Source String** **or** the **Destination String** is full.

When either of these things happen the next **Destination String** becomes the receiving area and characters are copied into it until it too is full or another **Delimiter** is encountered in the **Source String**.

Characters are copied from the **Source String** to the **Destination Strings** according to the rules for Alphanumeric moves. There **is** space filling.



UNSTRING Termination

The UNSTRING statement terminates when:-

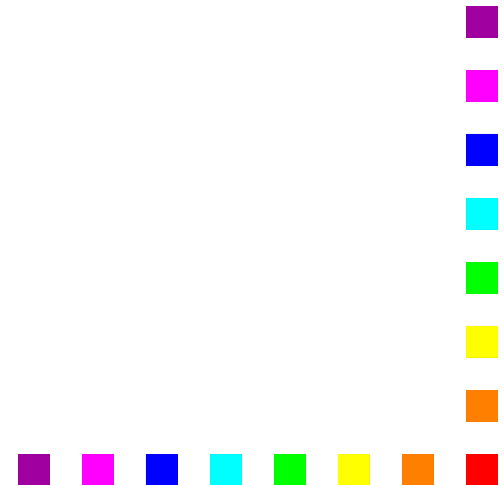
All the characters in the **Source String** have been **examined**

OR

All the **Destination Strings** have been **processed**

OR

Some **error** condition is **encountered**.



UNSTRING clauses.

ON OVERFLOW.

The ON OVERFLOW is activated if :-

The Unstring pointer (Pointer#i) is **not pointing** to a character position **within** the **SourceString** when the UNSTRING executes.

All the Destination Strings have been processed but there are still valid **unexamined characters in the Source String**.

COUNT IN

The COUNT IN clause is associated with a particular Destination String and holds a **count** of the number of **characters** passed to the Destination String.

TALLYING IN

Only one TALLYING clause can be used with each UNSTRING. It holds a **count** of the number of **Destination Strings** affected by the UNSTRING operation.



The UNSTRING clauses 2.

WITH POINTER

The **Pointer#i** holds the **position** of the next non-delimiter character to be **examined** in the **Source String**.

Pointer#i must be **large enough** to hold a value **one greater** than the size of the **Source String**.

DELIMITER IN

A **DELIMITER IN** clause is associated with a particular Destination String. **HoldDelim\$i** holds the **Delimiter** that was **encountered** in the **Source String**.

ALL

When the **ALL** phrase is used, **contiguous delimiters** are treated as if only **one delimiter** had been encountered.



UNSTRING Example 1

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 - 0 5 - 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

--

UNSTRING Example 1

01 DayStr PIC XX.

01 MonthStr PIC XX.

01 YearStr PIC XX.

01 DateStr PIC X(8).

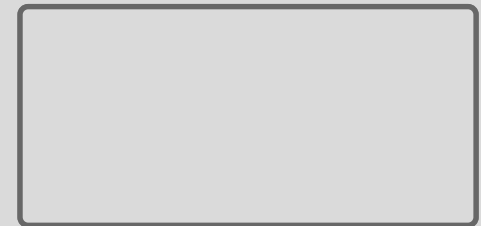
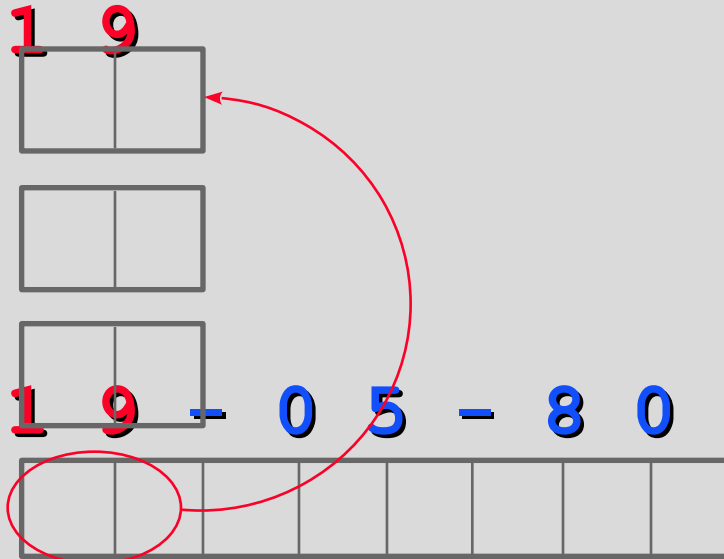
ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.



UNSTRING Example 1

01 DayStr PIC XX.

01 MonthStr PIC XX.

01 YearStr PIC XX.

01 DateStr PIC X(8).

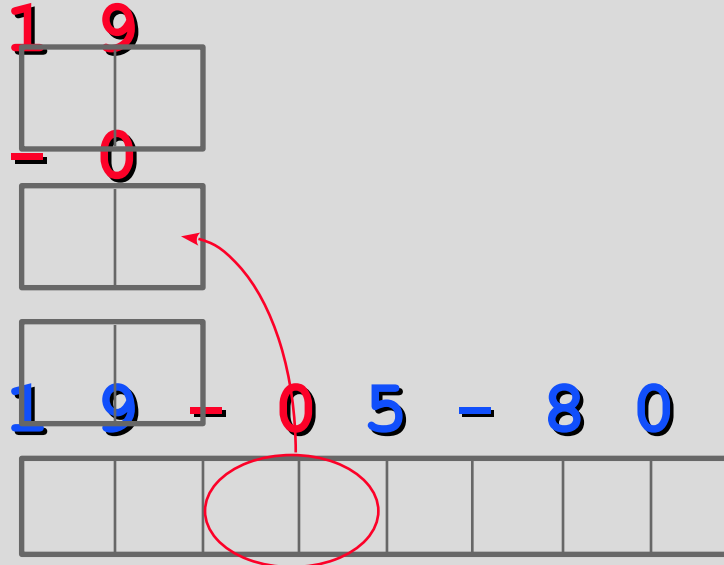
ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.



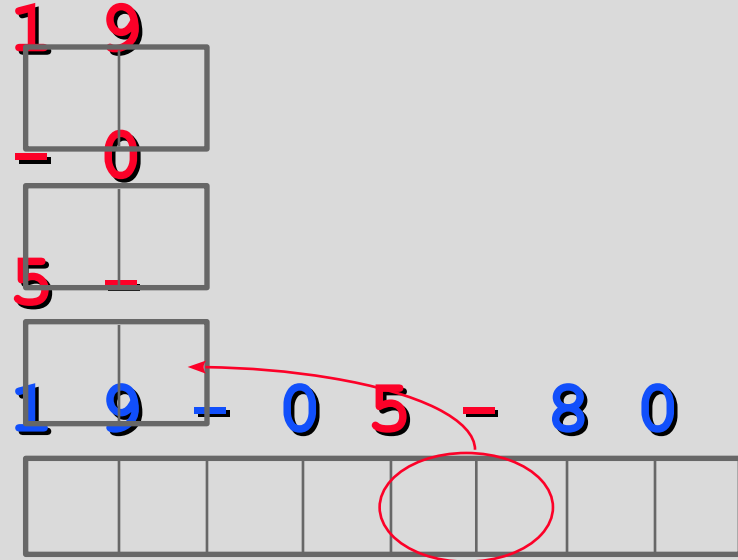
UNSTRING Example 1

01 DayStr PIC XX.

01 MonthStr PIC XX.

01 YearStr PIC XX.

01 DateStr PIC X(8).



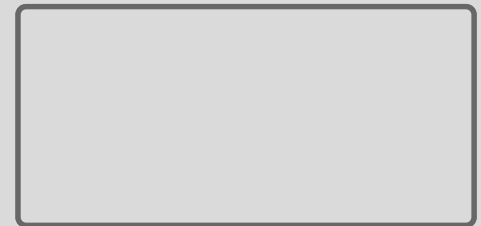
ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.



UNSTRING Example 1

01 DayStr PIC XX.

1 9
[][]

01 MonthStr PIC XX.

- 0
[][]

01 YearStr PIC XX.

5 -
[][]

01 DateStr PIC X(8).

1 9 - 0 5 - 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

Chars Left

UNSTRING Example 2

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 s t o p 0 5 s t o p 8 0

--	--	--	--	--	--	--	--	--	--	--	--	--	--

DATESTR.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.

--

UNSTRING Example 2

01 DayStr PIC XX.

1 9

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 s t o p 0 5 s t o p 8 0

--	--	--	--	--	--	--	--	--	--	--	--	--	--

MOVE DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.

--

UNSTRING Example 2

01 DayStr PIC XX.

1 9

--	--

01 MonthStr PIC XX.

0 5

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 s t o p 0 5 s t o p 8 0

--	--	--	--	--	--	--	--	--	--	--	--	--	--

MOVE DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.

--

UNSTRING Example 2

01 DayStr PIC XX.

1 9

--	--

01 MonthStr PIC XX.

0 5

--	--

01 YearStr PIC XX.

8 0

--	--

01 DateStr PIC X(8).

1 9 s t o p 0 5 s t o p 8 0

--	--	--	--	--	--	--	--	--	--	--	--	--	--

MOVE DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.

--

UNSTRING Example 3

01 DayStr PIC XX.

1 9

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 - 0 5 / 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE Ye

DISPLAY Hold1 SPACE Hold2

--

UNSTRING Example 3

01 DayStr PIC XX.

1	9

01 MonthStr PIC XX.

0	5

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 - 0 5 / 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE Ye

DISPLAY Hold1 SPACE Hold2

--	--	--	--	--	--	--	--

UNSTRING Example 3

01 DayStr PIC XX.

1 9
[][]

01 MonthStr PIC XX.

0 5
[][]

01 YearStr PIC XX.

8 0
[][]

01 DateStr PIC X(8).

1 9 - 0 5 / 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE Ye

DISPLAY Hold1 SPACE Hold2

--	--	--	--	--	--	--	--

UNSTRING Example 3

01 DayStr PIC XX.

1 9

--	--

01 MonthStr PIC XX.

0 5

--	--

01 YearStr PIC XX.

8 0

--	--

01 DateStr PIC X(8).

1 9 - 0 5 / 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE Ye

DISPLAY Hold1 SPACE Hold2

19 05 80

- /

UNSTRING Example 4

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1 9 - 0 5 / 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold1

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE Ye

DISPLAY Hold1

--

UNSTRING Example 4

01 DayStr PIC XX.

1 9

--	--

01 MonthStr PIC XX.

0 5

--	--

01 YearStr PIC XX.

8 0

--	--

01 DateStr PIC X(8).

1 9 - 0 5 / 8 0

--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold1

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE Ye

DISPLAY Hold1

19 05 80

/

UNSTRING Example 5

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(11).

1 5 - - - 0 7 - - 9 4

--	--	--	--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY ALL "-"

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

--

UNSTRING Example 5

01 DayStr PIC XX.

1 5

--	--

01 MonthStr PIC XX.

0 7

--	--

01 YearStr PIC XX.

9 4

--	--

01 DateStr PIC X(11).

1 5 - - - 0 7 - - 9 4

--	--	--	--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY ALL "-"

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

--

UNSTRING Example 6

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(11).

1 5 - - - 0 7 - - 9 4

--	--	--	--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "-"

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

--

UNSTRING Example 6

01 DayStr PIC XX.

01 MonthStr PIC XX.

01 YearStr PIC XX.

01 DateStr PIC X(11).

1 5

--	--

--	--

--	--

1 5 - - 0 7 - - 9 4

--	--	--	--	--	--	--	--	--	--	--	--

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "-"

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

Chars Left

UNSTRING Example 7

```
01 OldName PIC X(80) .
```

Tim John Roberts

```
01 TempName.
```

```
02 NameInitial PIC X.
```

```
02 FILLER PIC X(15) .
```

```
01 NewName PIC X(30) .
```

```
01 Pointers.
```

```
02 StrPtr PIC 99 VALUE 1.
```

```
02 UnstrPtr PIC 99 VALUE 1.
```

```
88 NameProcessed VALUE 81.
```

```
PROCEDURE DIVISION.
```

```
ProcessName.
```

```
ACCEPT OldName.
```

```
PERFORM UNTIL NameProcessed
```

```
UNSTRING OldName DELIMITED BY ALL SPACES
```

```
INTO TempName WITH POINTER UnstrPtr
```

```
END-UNSTRING
```

```
Display TempName
```

```
END-PERFORM
```

```
STOP RUN.
```

UNSTRING Example 7

```
01 OldName PIC X(80) .
```

Tim John Roberts

```
01 TempName.
```

T im

```
02 NameInitial PIC X.  
02 FILLER PIC X(15) .
```

```
01 NewName PIC X(30) .
```

```
01 Pointers.
```

```
02 StrPtr PIC 99 VALUE 1.  
02 UnstrPtr PIC 99 VALUE 1.  
88 NameProcessed VALUE 81.
```

Tim

```
PROCEDURE DIVISION.
```

```
ProcessName.
```

```
ACCEPT OldName.
```

```
PERFORM UNTIL NameProcessed
```

```
UNSTRING OldName DELIMITED BY ALL SPACES
```

```
INTO TempName WITH POINTER UnstrPtr
```

```
END-UNSTRING
```

```
Display TempName
```

```
END-PERFORM
```

```
STOP RUN.
```

UNSTRING Example 7

```
01 OldName    PIC X(80) .
```

Tim John Roberts

```
01 TempName.
```

```
02 NameInitial PIC X.
```

```
02 FILLER      PIC X(15) .
```

J ohn

```
01 NewName          PIC X(30) .
```

```
01 Pointers.
```

```
02 StrPtr          PIC 99 VALUE 1.
```

```
02 UnstrPtr        PIC 99 VALUE 1.
```

```
88 NameProcessed VALUE 81.
```

John

```
PROCEDURE DIVISION.
```

```
ProcessName.
```

```
ACCEPT OldName.
```

```
PERFORM UNTIL NameProcessed
```

```
    UNSTRING OldName DELIMITED BY ALL SPACES
```

```
        INTO TempName WITH POINTER UnstrPtr
```

```
    END-UNSTRING
```

```
        Display TempName
```

```
END-PERFORM
```

```
STOP RUN.
```

UNSTRING Example 7

```
01 OldName PIC X(80) .
```

Tim John Roberts 

```
01 TempName.
```

R oberts 

```
02 NameInitial PIC X.  
02 FILLER PIC X(15) .
```

```
01 NewName PIC X(30) .
```

```
01 Pointers.
```

```
02 StrPtr PIC 99 VALUE 1.  
02 UnstrPtr PIC 99 VALUE 1.  
88 NameProcessed VALUE 81.
```

Roberts

```
PROCEDURE DIVISION.
```

```
ProcessName.
```

```
ACCEPT OldName.
```

```
PERFORM UNTIL NameProcessed
```

```
UNSTRING OldName DELIMITED BY ALL SPACES
```

```
INTO TempName WITH POINTER UnstrPtr
```

```
END-UNSTRING
```

```
Display TempName
```

```
END-PERFORM
```

```
STOP RUN.
```

UNSTRING Example 8

OldName **Tim John Roberts**



TempName

NewName



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

PERFORM UNTIL NameProcessed
STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr
END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

END-PERFORM
STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING
STOP RUN.

UNSTRING Example 8

OldName

Tim John Roberts

TempName

T im

NewName

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr
END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr
END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim John Roberts**



TempName **T**im

NewName **T.**



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr
END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John**  **Roberts**



TempName **J** **ohn**

NewName **T** .



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John** **Roberts**



TempName **J** **ohn**

NewName **T** **.** **J** **.**



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr
END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John** **Roberts** 

TempName **R** **oberts**

NewName **T.J.** 



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

**UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr**

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim John Roberts**

TempName **R oberts**

NewName **T . J . Roberts**

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.